Relative Clause Extraposition in English: An Information-theoretic Approach

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Relative clause extraposition (RCE) in English (Table 1), where a relative clause is displaced rightward, involves a cross-linguistically rare 'non-projective dependency' and imposes greater localized processing costs compared to the adjacent order [1]. Why should speakers use a complex structure like RCE when a simpler structure is available? Previous studies have focused on grammatical weight and discourse factors, showing that speakers use RCE more often when it allows for short-before-long constituent order and when it expresses discourse focus on the subject [2-4]. From an information-theoretic perspective, the Uniform Information Density (UID) hypothesis [5-7], which proposes that speakers tend to choose forms which allow for a uniform distribution of information across an utterance, may provide additional insights into speakers' structural choices. This hypothesis has been successful in accounting for certain alternations such as optional *that* in English [4]. In addition, it has been successfully applied to help explain patterns of RCE and adjacent order in historical German [8]. When examining several word order alternations in English, however, minimal support was found [9]. Importantly, different studies have defined and implemented the information density in different ways. The current pilot study investigates the structural choice between adjacent and extraposed RCs in English, focusing on how speakers weigh different linguistic factors. More specifically, we aim to address two questions: 1) How do adjacent and extraposed RCs differ in their information-theoretic profiles (i.e., surprisal-based measures)? 2) Do information-theoretic factors improve the machine learning model's prediction accuracy for speakers' RC choices beyond grammatical weight and discourse factors?

Methods 390 sentences (332 adjacent RCs, 58 extraposed RCs) from the ICE-GB corpus (the British component of the International Corpus of English) and their annotations for weight-related and discourse-related factors were obtained from the authors of Francis and Michaelis (2014). For the information-theoretic factors, we first computed word-level surprisal values, operationalized as the negative log-likelihood of a word given its previous context [10] as calculated by DistilGPT2 [11], and then derived two sentence-level measures: mean surprisal and rolling surprisal variance, which reflect overall information density and the uniformity of information distribution, respectively. Table 2 provides definitions of all factors. To address RQ1, we used Mann–Whitney U tests to examine whether surprisal-based differences between adjacent and extraposed sentences were statistically significant. To address RQ2, we trained machine-learning classifiers to model speakers' structural choices (adjacent vs. extraposed RC) using two different sets of predictors: (a) a baseline model including only weight- and discourse-based factors, and (b) a full model that additionally incorporates information-theoretic factors. Comparing classification performance allowed us to evaluate the added predictive value of information-theoretic measures. Model interpretability was further assessed using SHAP analysis [12] to quantify the contributions of individual predictors to the classifier's decisions

Results & Discussion For RQ1, we found that extraposed RCs showed higher mean surprisal (extraposed: Mean= 5.83, Std = 1.11; adjacent: Mean = 5.61, Std= 1.07), and higher rolling surprisal variance (extraposed: M = 9.69, SD = 7.13; adjacent: M = 8.51, SD = 5.87) than adjacent RCs, but neither difference reached significance ($p > .05$). For RQ2, gradient boosting classifier (Table 3) showed strong performance in predicting speakers' choices, and adding information-theoretic factors improved classification performance (e.g., F1 score: 0.902→0.917). SHAP analyses (Figure 1) further showed that VP-to-RC length ratio and VP length were the most influential predictors, while mean surprisal and surprisal variance still outperformed several discourse-related factors. In conclusion, information-theoretic factors appear to capture additional dimensions of information flow and create a fuller picture of how RCE enhances communicative efficiency [13]. The current study provides proof-of-concept for a larger dissertation study, in which we plan to significantly enlarge the dataset and examine additional metrics related to information density, discourse genre, and language background.

Table 1 Corpus example of extraposed RC and constructed example of its adjacent counterpart

Adjacent RC	**A series of changes** which are histologically described as Wallerian degeneration **occurs**
Extraposed RC	**A series of changes occurs** which are histologically described as Wallerian degeneration.

Table 2 Overview of the independent factors used in the sentence annotation

categories	factors
weight-based factors	- subj_np_len: the number of words in the subject position - vp_len: the number of words in verb phrase(VP) position - rc_len: the number of words in relative clause(RC) position - vp_to_rc_ratio: Ratio of VP to RC length - vp_minus_rc_len: VP length minus RC length - subj_to_rc_ratio: subject NP-to-extraposed RC length ratio
discourse-based factors	- pred_type: predicate types (unaccusative(non-copular), unergative, copular (pred nom, pred PP, pred adjective), transitive - definiteness: definiteness of the subject NP (definite, indefinite) - subj_head_type: lexical=L, pronominal=P - subj_accessibility: subject accessibility (No prior mention (new), Superset mention (sm), Prior mention (given)) - pred_accessibility: predicate accessibility ((No prior mention (new), Superset mention (sm), Prior mention (given))
information-theoretic factors	- mean_surp: surprisal value of each word summed and divided by the number of words - rolling_surp_var: the variance of the rolling mean surprisal across adjacent word pairs in the sentence

Table 3: Evaluation metrics of the machine learning classifier in predicting RCE choice with three different factor sets. Class 0 /negative class is adjacent RC; Class 1 /positive class is extraposed RC. All evaluation metric values are weighted.

Gradient boosting classifier	**a) baseline model** (only weight- and discourse-based factors)	**b) full model** (weight, discourse, and information-theoretic factors)	c) model with information-theoretic factors only
accuracy	0.910	0.923	0.744
precision	0.920	0.930	0.587
recall	0.910	0.923	0.744
F1 score	0.902	0.917	0.656

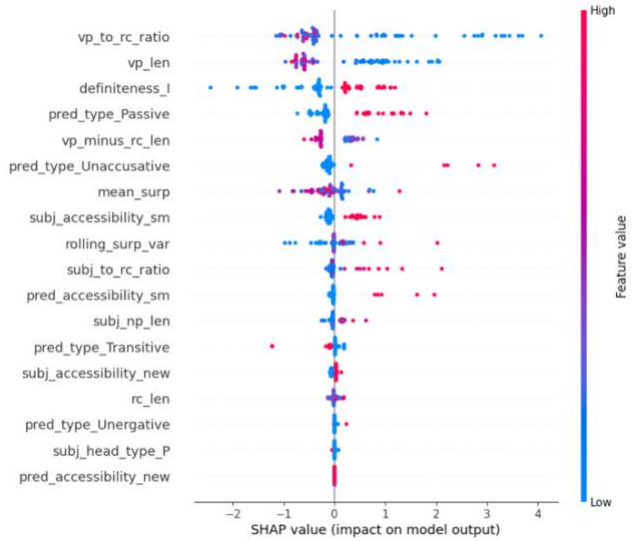

Figure 1 SHAP summary plot showing the relative importance and direction of linguistic factors in predicting RCE choice. Each dot shows how important that feature was for one individual prediction. Positive SHAP values indicate a higher likelihood of extraposition. Feature values are color-coded (red = high, blue = low).

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